

Experiment No. 7

Title:

Demonstration of Dynamic DOM Manipulation and Event Handling; Development of an Interactive To-Do List Application.

Software / Tools Required:

1. Visual Studio Code (VS Code)
2. Google Chrome
3. HTML5
4. CSS3
5. JavaScript (ES6)

Theory:

The Document Object Model (DOM) represents an HTML document as a tree of objects. JavaScript can be used to create, modify, and remove HTML elements dynamically.

(a) DOM Manipulation

DOM manipulation allows JavaScript to change the content and structure of a webpage.

(b) Event Handling

Event handling allows JavaScript to respond to user actions such as clicking buttons, entering text, and submitting forms.

(c) Creating Elements

The `document.createElement()` method is used to create new HTML elements dynamically.

(d) Replacing Elements

The `replaceChild()` method is used to replace an existing element with another element.

(e) Removing Elements

The `remove()` method is used to remove an element from the DOM.

To-Do List Application

Accepts User Input Such As:

1. Task Name

It Performs:

1. Adds new tasks dynamically.
2. Displays tasks in a list.
3. Allows tasks to be edited.
4. Saves edited tasks.
5. Deletes tasks.
6. Prevents empty task submission.

Event-Driven Registration Form

The practical also demonstrates form validation using events such as **focus**, **blur**, **change**, and **submit**. These events provide real-time feedback and validate user-entered information.

Result / Conclusion:

The practical was completed successfully. The concepts of **DOM manipulation and event handling** were implemented. An interactive To-Do List application was developed using JavaScript that allows users to add, edit, and delete tasks dynamically. Form validation and event handling were also demonstrated successfully.